

FACULTY OF ELECTRICAL ENGINEERING, AUTOMATICS, COMPUTER SCIENCE AND BIOMEDICAL ENGINEERING

Design of a Gripper for an Industrial Manipulator
Gripper No. 15, type P-(P-O-O)

Maria Malinka

Automatic Control and Robotics

Tutorial Group No. 3

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1. Design task

Design a gripper for an industrial manipulator, following the given kinematic scheme, that meets the following requirements:

- during transport, the gripping device must pick up (grip) the object at its initial position, hold it for the whole transport operation, and release it at the target location;
- the transported objects are steel shafts with diameter **d = 78 mm** and length **l = 100 mm**;
- the manipulator is supplied with compressed air at a nominal pressure of **p_n = 0.6 MPa**.

2. Mobility (degrees of freedom) of the gripper

The calculation uses the following formula:

$$W = 3n - p_4 - 2p_5$$

where:

- n – number of moving links,
- p_4 – number of class-4 kinematic pairs,
- p_5 – number of class-5 kinematic pairs.

The following kinematic scheme was assigned for this project:

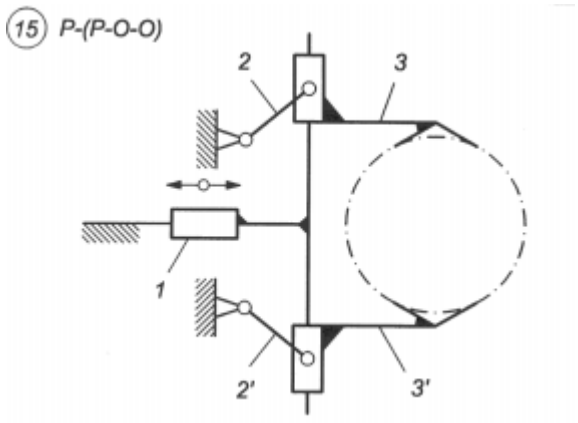


Fig. 1: Kinematic scheme of the gripper

For this kinematic scheme:

- Moving links: (1, 2, 2′, 3, 3′), so **n = 5**.
- Class-4 kinematic pairs: none, so **p_4 = 0**.
- Class-5 kinematic pairs: (0, 1), (0, 2), (0, 2′), (1, 3), (1, 3′), (2, 3), (2′, 3′), so **p_5 = 7**.

Using the mobility formula (2.1):

$$W = 3 \cdot 5 - 0 - 2 \cdot 7 = 1$$

The mobility equals 1, which means a single linear pneumatic actuator is enough for the gripper to work correctly.

3. Analysis of the design task, list of requirements and computational model

3.1. Actuator stroke, element dimensions, jaw range, and size and weight of the manipulated object

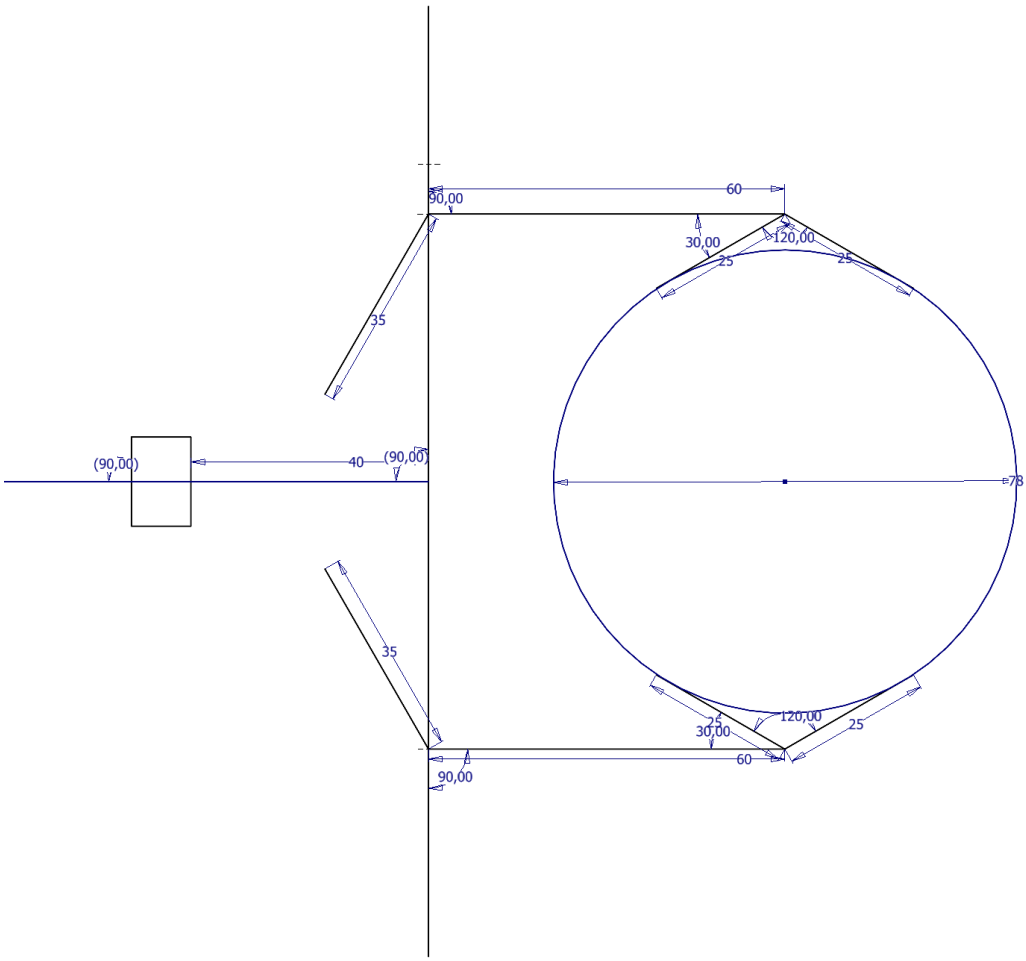


Fig. 2: Gripper model designed in Inventor Professional 2026

I assume a maximum shaft diameter of 78 mm and a minimum shaft diameter of 62 mm. For these dimensions the actuator stroke is approximately **Δx = 10 mm**.

The maximum weight of the transported object was determined with:

$$Q_{max} = \frac{\pi d^2}{4} l \gamma$$

where:

- d – shaft diameter, 78 mm,
- l – shaft length, 100 mm,
- γ – specific weight of the shaft material (steel), 78 000 N/m³.

Substituting into (3.1):

$$Q_{max} \approx 37.27\,\mathrm{N}$$

3.2. Maximum required gripping force

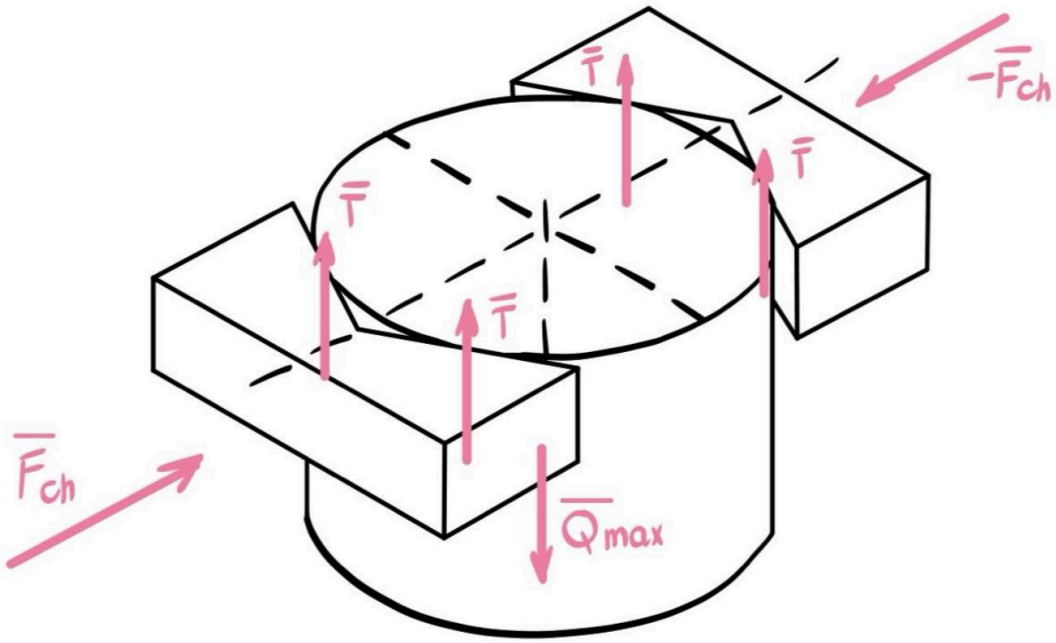


Fig. 3: Distribution of friction forces while gripping the object

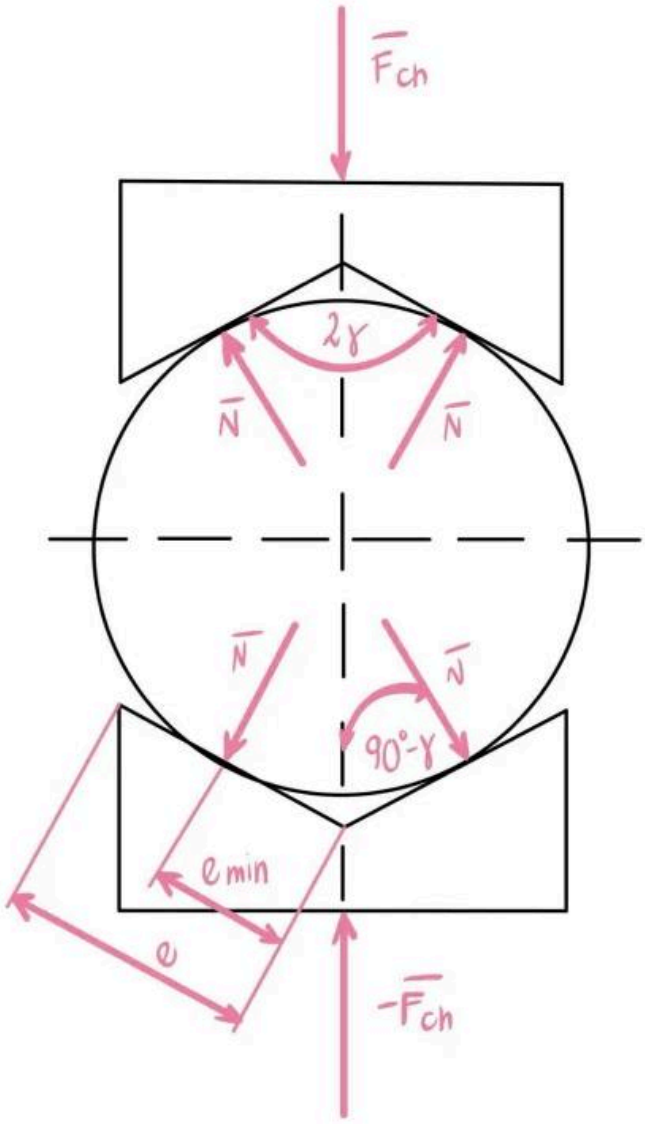


Fig. 4: Distribution of normal forces while gripping the object

Coefficients adopted for further calculations:

- gripper overload factor: **n = 2**,
- coefficient of friction (steel–steel): **μ = 0.15**,
- jaw opening angle: **2γ = 120°**.

For the transported element to be gripped correctly, the following condition (from the geometry of Fig. 3) must be satisfied:

$$4T \geq Qn$$

Taking the jaw opening angle into account, the gripping force is:

$$F_{ch} = 2N \cos\ (\ 90^{\circ} - \gamma\ )\ = 2N \sin\ \gamma$$

So:

$$N = \frac{F_{ch}}{2 \sin \gamma}$$

Multiplying both sides by the coefficient of friction gives the friction force:

$$T = \mu N = \frac{\mu F_{ch}}{2 \sin \gamma}$$

Using inequality (3.2):

$$\frac{4 \mu F_{ch}}{2 \sin \gamma} \geq Qn$$

The final formula is:

$$F_{ch} \geq \frac{Qn \sin \gamma}{2 \mu}$$

Hence the gripping force:

$$F_{ch} \geq \frac{37.27 \cdot 2 \cdot \sin\left(60^\circ\right)}{2 \cdot 0.15} \approx 215.2\,\text{N}$$

For further calculations I adopt **F_ch = 220 N**.

3.3. Jaw dimensions

Assuming the shaft contacts each jaw at two points, the necessary condition for correct gripping is:

$$e \geq e_{min} = \frac{d_{max}}{2 \tan \gamma}$$

Substituting the data:

$$e \geq e_{min} = \frac{78}{2 \tan\left(60^\circ\right)} \approx 22.52\,\text{mm}$$

The minimum value is therefore consistent with the adopted jaw dimension **e = 25 mm**.

4. Displacement characteristic

With a suitable scheme drawn, the relationship y(x) can be derived.

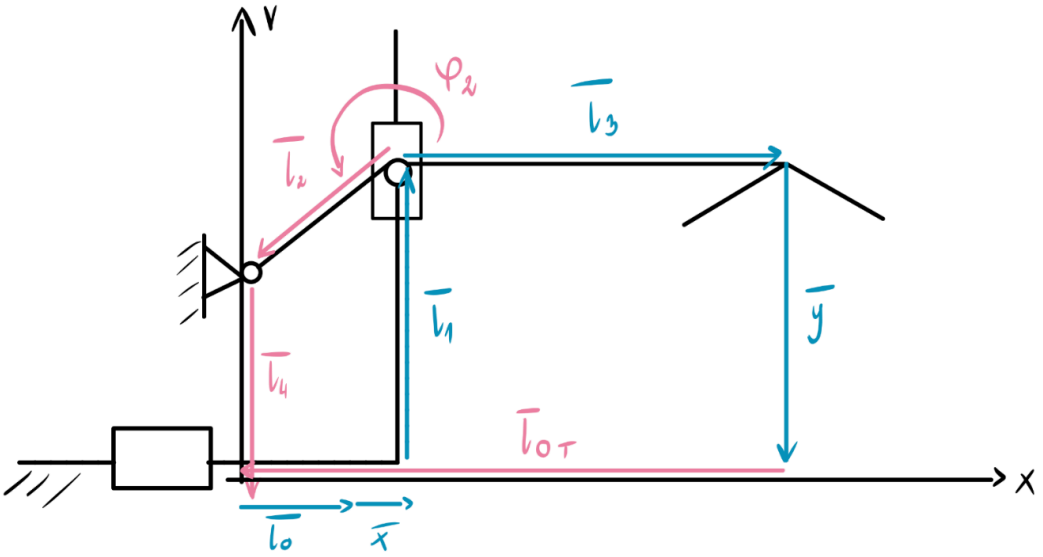


Fig. 5: Scheme used to derive the relationship y(x); the gripper arms are fully open

Vector polygon equation:

$$\bar{x} + \bar{l_1} + \bar{l_2} + \bar{l_4} + \bar{l_0} = 0$$

Projecting the vectors onto the OX axis:

$$x + l_2 \cos \varphi_2 + l_0 = 0$$

so:

$$\cos \varphi_2 = -\frac{l_0 + x}{l_2}$$

Projecting onto the OY axis:

$$l_1 + l_2 \sin \varphi_2 - l_4 = 0$$

so:

$$l_1 = l_4 - l_2 \sin \varphi_2$$

Since the length l_1 corresponds to y(x):

$$y = l_4 - l_2 \sin \varphi_2$$

Using the trigonometric identity $\sin \varphi_2 = \pm \sqrt{1 - \cos^2 \varphi_2}$, and reading from the geometry of the figure that the sine is negative, we transform:

$$y = l_4 + l_2 \sqrt{1 - \cos^2 \varphi_2}$$

$$y = l_4 + l_2 \sqrt{1 - \left(\frac{l_0 + x}{l_2}\right)^2}$$

The final formula is:

$$y = l_4 + \sqrt{l_2^2 - \left(l_0 + x \right)^2}$$

For this gripper we adopt:

- l_0 = 17.5 mm
- l_4 = 14.5 mm
- l_2 = 35 mm

Substituting into (4.4):

$$y = 14.5 + \sqrt{35^2 - \left(17.5 + x \right)^2}$$

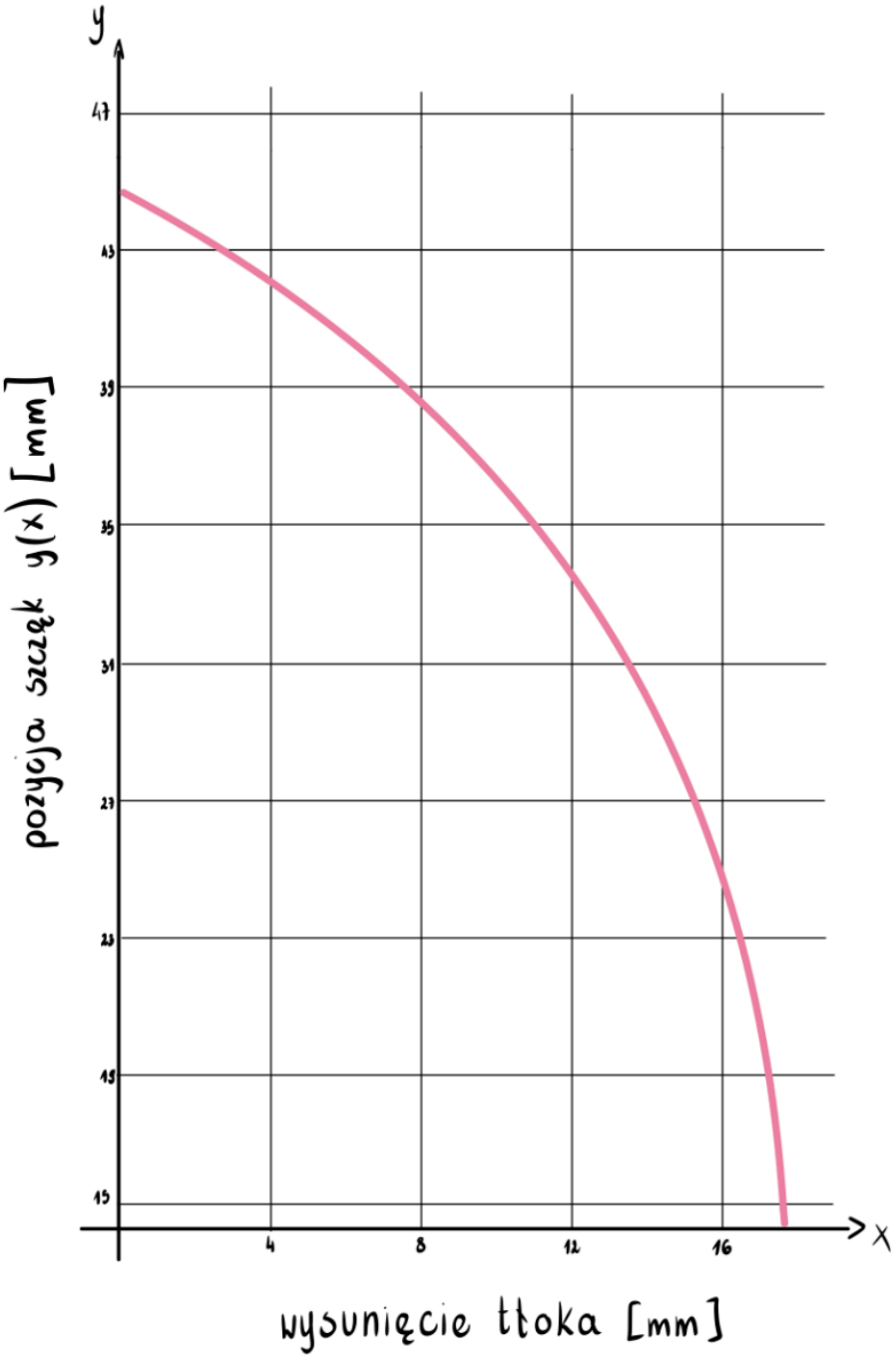


Fig. 6: Displacement characteristic y(x). Vertical axis: jaw position y(x) [mm]; horizontal axis: piston extension [mm].

5. Velocity characteristic

To determine the velocity characteristic, the time derivative of relationship (4.4) was calculated:

$$f_v(x) = \frac{d}{dt}y(x) = \frac{d}{dx}\left(l_4 + \sqrt{l_2^2 - (l_0 + x)^2}\right) = -\frac{l_0 + x}{\sqrt{l_2^2 - (l_0 + x)^2}}$$

Substituting the data:

$$f_v(x) = -\frac{17.5 + x}{\sqrt{35^2 - (17.5 + x)^2}}$$

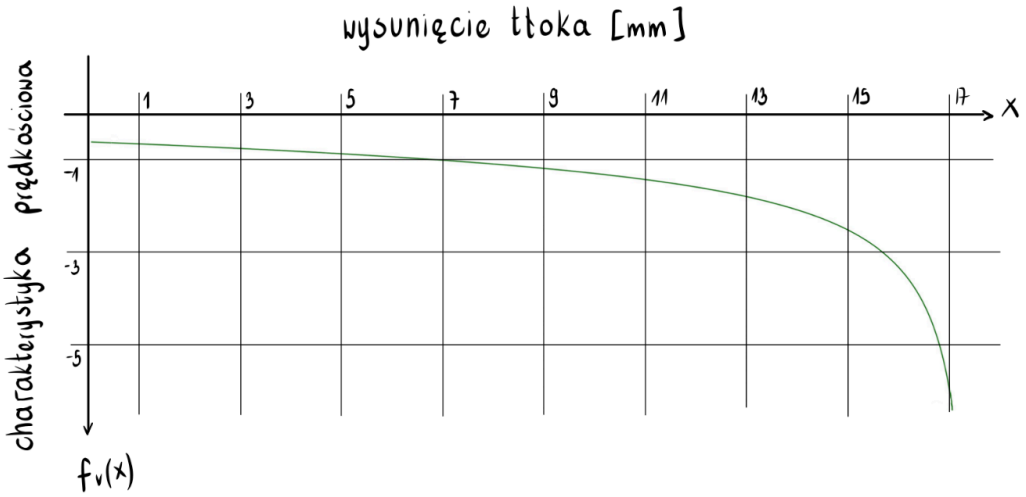


Fig. 7: Velocity characteristic f_v(x). Horizontal axis: piston extension [mm]; vertical axis: velocity characteristic.

6. Force characteristic

The force characteristic is defined as:

$$f_F(x) = \frac{F_{ch}}{F_s}$$

where:

- *F_ch* – gripping force,
- *F_s* – force at the output of the drive unit.

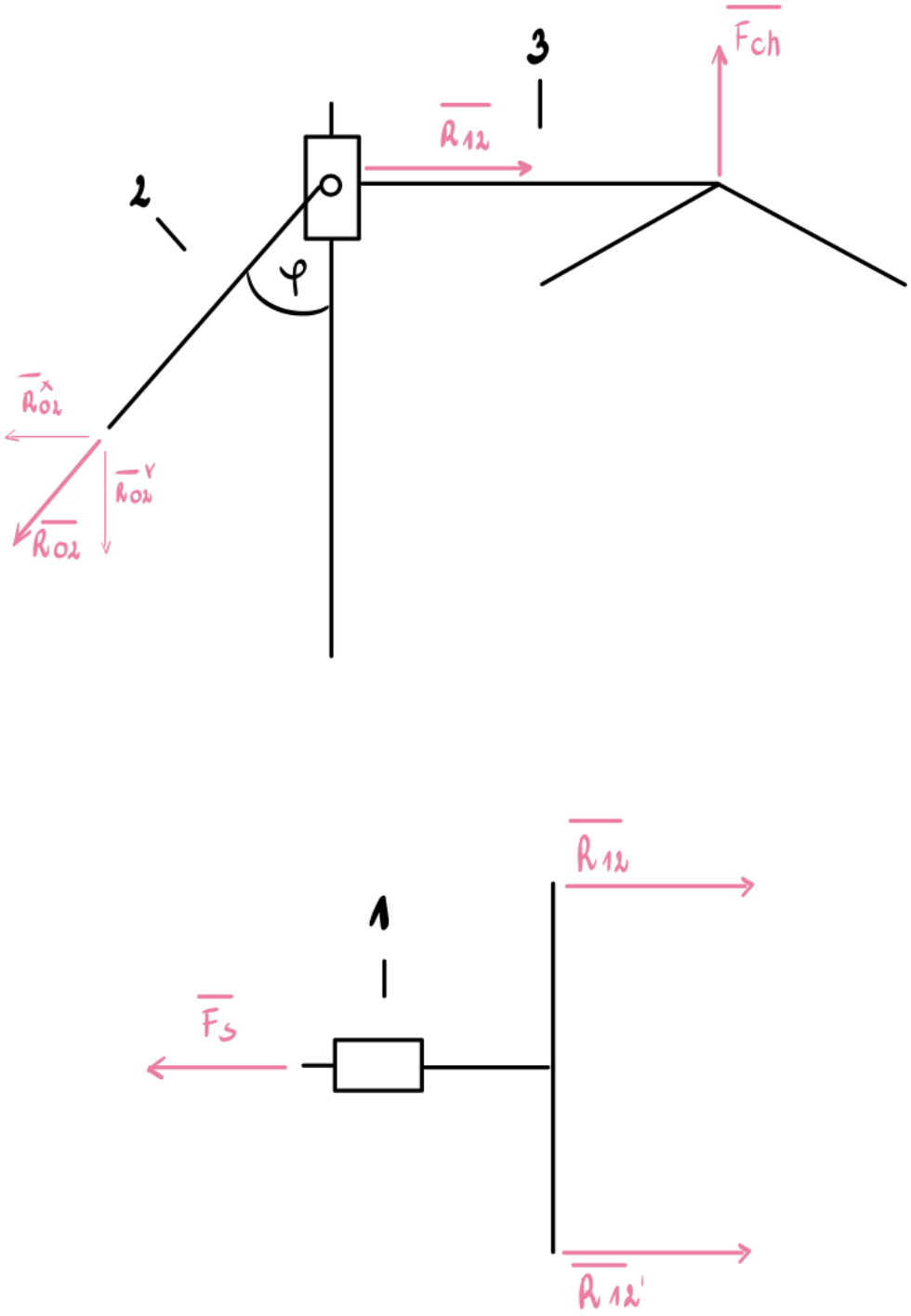


Fig. 8: Scheme used to derive f_F(x), force distribution

Equilibrium condition for group (2, 3):

$$\bar{R_{02}} + \bar{R_{12}} + \bar{F_{ch}} = 0$$

Projecting onto the OX and OY axes:

$$\begin{cases} R_{12} - R_{02} \sin \varphi = 0 \\ F_{ch} - R_{02} \cos \varphi = 0 \end{cases}$$

Equilibrium condition for element 1:

$$\bar{F}_s + \bar{R}_{12} + \bar{R}_{12'} = 0$$

Projected onto the OX axis:

$$F_s - R_{12} - R_{12'} = 0$$

By symmetry of element 1:

$$R_{12} = R_{12'}$$

Combining all the conditions above:

$$F_{ch} = R_{02} \cos \varphi$$
$$F_s = 2R_{02} \sin \varphi$$

Substituting into (6.1):

$$f_F(x) = \frac{F_{ch}}{F_s} = \frac{R_{02} \cos \varphi}{2R_{02} \sin \varphi} = \frac{\cos \varphi}{2 \sin \varphi} = \frac{1}{2} \cot \varphi$$

From the geometry of the system:

$$\cot \varphi = \frac{\sqrt{l_2^2 - (l_0 + x)^2}}{l_0 + x}$$

Combining (6.4) and (6.5) gives the final relationship:

$$f_F(x) = \frac{1}{2} \frac{\sqrt{l_2^2 - (l_0 + x)^2}}{l_0 + x}$$

Substituting the data:

$$f_F(x) = \frac{1}{2} \frac{\sqrt{35^2 - (17.5 + x)^2}}{17.5 + x}$$

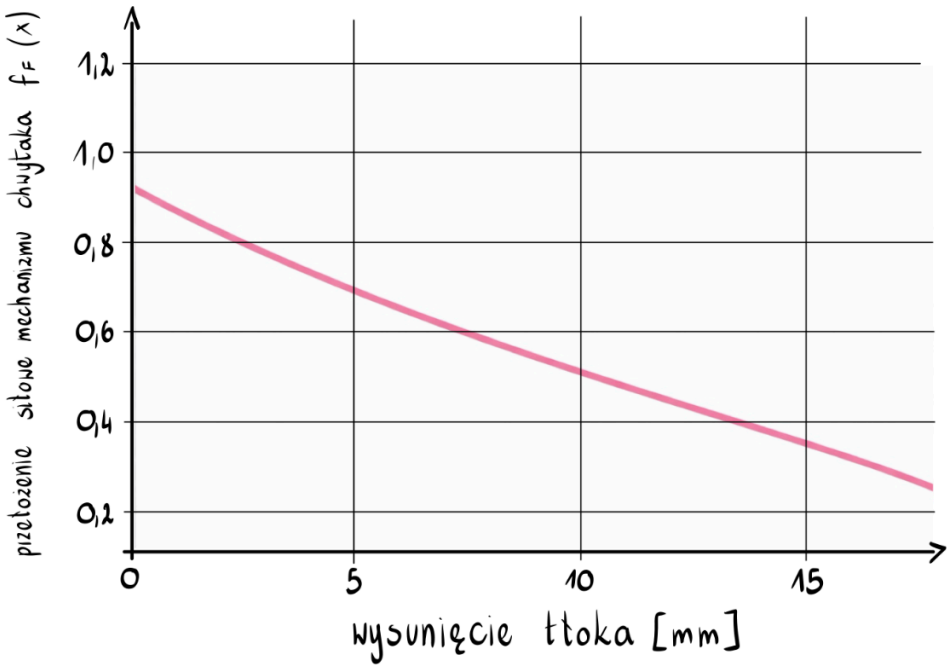


Fig. 9: Force characteristic of the gripper f_F(x). Vertical axis: force ratio of the gripper mechanism; horizontal axis: piston extension [mm].

7. Verifying the force characteristic with the instantaneous power method

The formula was verified with the instantaneous power method, which equates the input and output power of the mechanism and obtains the force ratio from the ratio of velocities. This lets us reuse the velocity characteristic derived earlier.

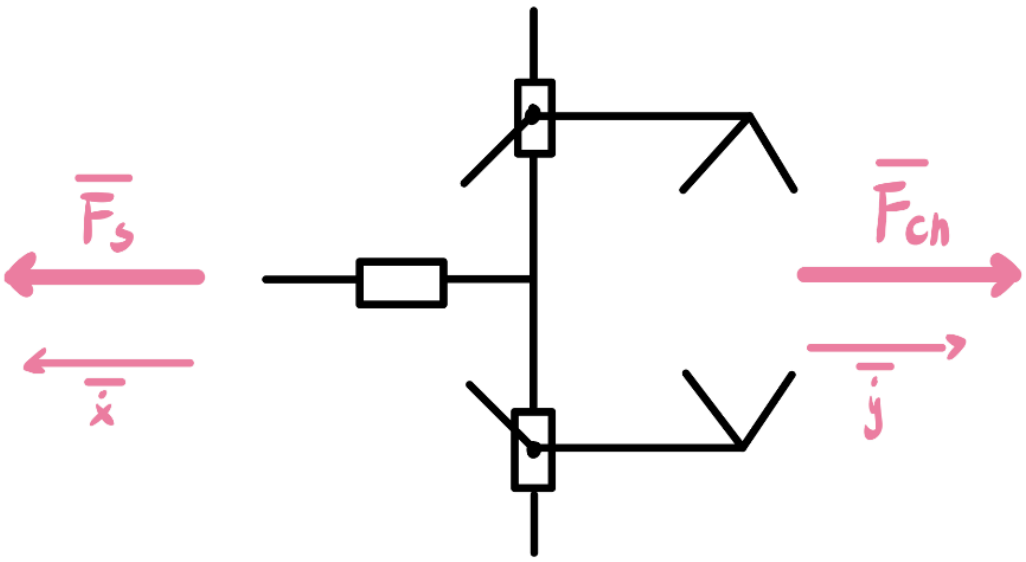


Fig. 10: Auxiliary scheme for applying the instantaneous power method to the gripper

Input power is assumed equal to output power:

$$P_{in} = P_{out}$$

So:

$$F_s \dot{x} = 2F_{ch} \dot{y}$$

Rearranging (7.2):

$$f_F(x) = \frac{F_{ch}}{F_s} = \frac{\dot{x}}{2\dot{y}} = \frac{1}{2f_v(x)} = -\frac{1}{2}\frac{\sqrt{l_2^2 - (l_0 + x)^2}}{l_0 + x}$$

The result from the instantaneous power method agrees with the earlier derivation. The difference in sign comes only from the assumed direction of displacement.

8. Maximum force required to drive the gripper

Based on the earlier assumptions, the maximum actuator stroke was set to 10 mm. The calculations show that this value ensures correct gripper operation while maintaining the required gripping force. A larger stroke reduces the force ratio of the mechanism and therefore increases the required drive force, so the smaller stroke was chosen as it gives more favourable working conditions.

For our stroke range, the graph (Fig. 9) shows that the force ratio is smallest at x = 10 mm, so this is the point at which the required pneumatic actuator force must be calculated.

From (6.6):

$$f_F(10) = 0.3936$$

Rearranging (6.1) gives the maximum force required at the driving link:

$$F_{s\,max} = \frac{F_{ch}}{f_F(x)} = \frac{220}{0.3936}\,\mathrm{N} = 558.9\,\mathrm{N}$$

9. Strength calculations

9.1. Shear strength condition for the most heavily loaded pin

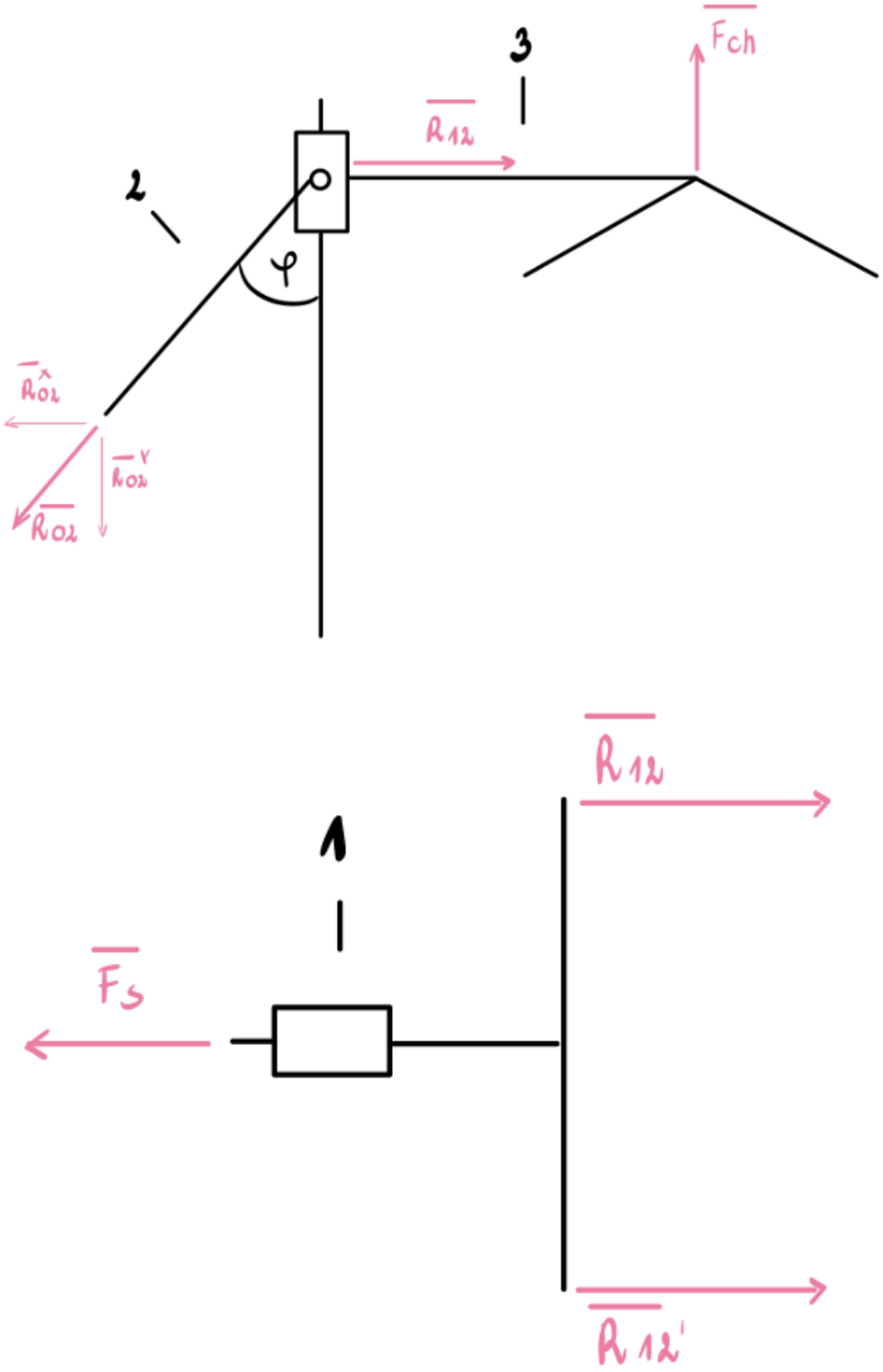


Fig. 11: Forces acting on the elements of the mechanism

From the equilibrium equations derived from the figure and used for the force characteristic:

$$R_{02} = \frac{F_{ch}}{\cos\varphi}, \qquad R_{12} = R_{02}\sin\varphi, \qquad F_s = 2R_{02}\sin\varphi, \qquad R_{02} = \frac{R_{12}}{\sin\varphi}$$

Since sin φ ≤ 1 and cos φ ≤ 1, these relationships show that the largest shear load on a pin occurs at the joint connecting elements 0 and 2. The reaction R_{02} was therefore used for the pin strength calculation.

The shear strength condition for the pin is:

$$\tau = \frac{F_t}{nA} \leq k_t$$

where:

- τ – shear stress,
- F_t – shear force,
- n – number of shear planes; here n = 2,
- A – pin cross-sectional area,
- k_t – allowable shear stress of the material; for S235 structural steel, with a safety margin, k_t = 80 MPa.

The pin has a circular cross-section:

$$A = \frac{\pi d^2}{4}$$

Following the reasoning above, the shear force used for the pin calculation equals the reaction R_{02}:

$$F_t = R_{02} = \frac{F_{ch}}{\cos\varphi} = \frac{220}{\cos 30^\circ} = 254\,\mathrm{N}$$

Rearranging the formulas above gives the minimum pin diameter:

$$d_{min} = 2\sqrt{\frac{F_t}{n\pi k_t}}$$

Substituting the data:

$$d_{min} = 2\sqrt{\frac{220}{2\cdot\pi\cdot 8\cdot 10^7}} = 1.32\cdot 10^{-3}\,\mathrm{m} = 1.32\,\mathrm{mm}$$

Because a standardised diameter must be used and a safety margin ensured, a pin with diameter **d = 4 mm** was selected.

Rearranging (9.1) and (9.2) gives:

$$\tau = \frac{4\cdot 220}{2\cdot\pi^2\cdot 0.004^2} = 2\,786\,332.6\,\mathrm{Pa} = 2.79\,\mathrm{MPa}$$

The shear stress in the pin is τ = 2.79 MPa, much lower than the allowable stress k_t = 80 MPa. **The strength condition is satisfied.**

9.2. Bending strength condition for the gripper arms

For the gripper arm we adopt the condition:

$$\sigma_g = \frac{M_g}{W_g} \leq k_g$$

where:

- σ_g – bending stress,
- M_g – bending moment,
- W_g – section modulus in bending,
- k_g – allowable bending stress of the material.

The bending moment is:

$$M_g = F_{ch}\, l$$

where F_ch = 220 N and l = 60 mm = 0.06 m.

Substituting into (9.8):

$$M_g = 220 \cdot 0.06 = 13.2\ \mathrm{Nm}$$

The section modulus is:

$$W_g = \frac{bh^2}{6}$$

with arm cross-section dimensions b = 0.01 m and h = 0.015 m. Therefore:

$$W_g = \frac{0.01 \cdot 0.015^2}{6} = 3.75 \cdot 10^{-7}\ \mathrm{m}^3$$

From (9.7) the bending stress is:

$$\sigma_g = \frac{13.2}{3.75 \cdot 10^{-7}} = 35.2 \cdot 10^6\ \mathrm{Pa} = 35.2\ \mathrm{MPa}$$

For S235 steel the maximum stress is about 235 MPa. Taking a safety factor into account, k_g = 140 MPa was adopted.

The calculated bending stress σ_g = 35.2 MPa is much lower than the allowable value, so the element works safely.

10. Minimum piston diameter of the drive cylinder

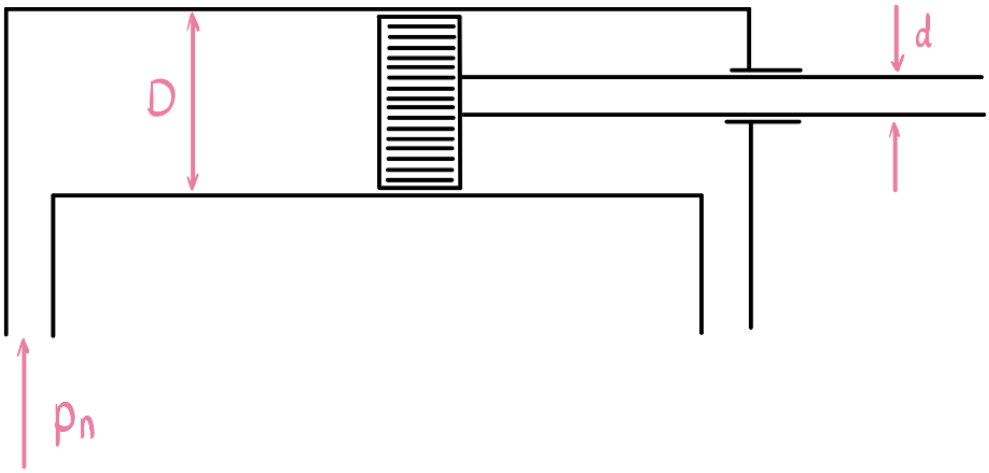


Fig. 12: Model of a double-acting pneumatic cylinder

The previously determined maximum required cylinder force is:

$$F_{s\,max} = 558.9\ \mathrm{N}$$

The nominal pressure is:

$$p_n = 0.6\ \mathrm{MPa}$$

To account for losses and non-ideal operating conditions, an overload factor k = 1.2 is used. Therefore:

$$F_w = kF_{s\,max}$$

Cylinder force as a function of pressure and piston area:

$$F_w = \frac{\pi D^2}{4} p_n$$

Rearranging gives the minimum piston diameter:

$$D_{min} = \sqrt{\frac{4kF_{s\,max}}{\pi p_n}}$$

Substituting the data:

$$D_{min} = \sqrt{\frac{4 \cdot 1.2 \cdot 558.9}{\pi \cdot 600000}} = 0.0377\ \mathrm{m}$$

A cylinder with diameter **D ≥ 37.7 mm** must therefore be selected.

11. Cylinder selection

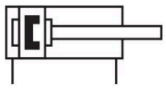
Based on the calculated minimum piston diameter D_min = 37.7 mm, a pneumatic cylinder with diameter D = 40 mm was selected. With the assumed stroke Δx = 10

mm, the **Festo ADN-S-40-10-A-P-A compact cylinder** (product no. 8092067) was chosen from the catalogue. It generates up to 754 N at 0.6 MPa, which meets the design requirements.

Siłownik kompaktowy
ADN-S-40-10-A-P-A
Numer produktu: 8092067

FESTO





General operating condition

Karta danych

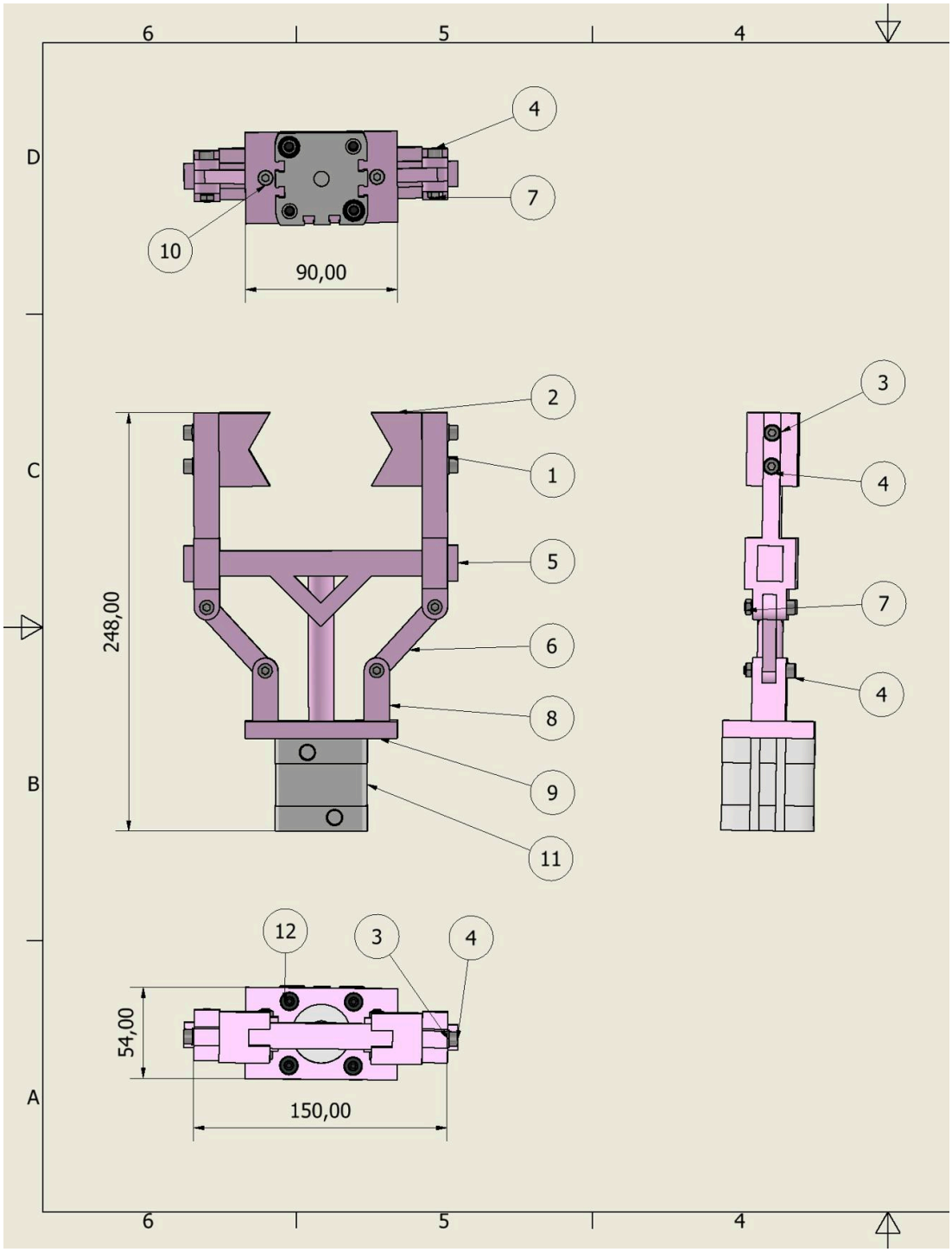
Cechy	Wartość
Skok	10 mm
Ø tłoka	40 mm
Amortyzacja	elastyczne pierścienie/płytki amortyzujące z obu stron
Pozycja montażu	dowolny
Sposób działania	dwustronnego działania
Zakończenie tłoczyska	Gwint zewnętrzny
Konstrukcja	Tłok Tłoczysko
Sygnalizacja położenia	do wyłącznika zbliżeniowego
Symbol	00991217
Warianty	Jednostronne tłoczysko
Ciśnienie robocze	0.06 MPa ... 1 MPa
Ciśnienie robocze	0.6 bar ... 10 bar
Medium robocze	Sprężone powietrze wg ISO 8573-1:2010 [7:4:4]
Uwaga dotycząca medium roboczego/sterującego	Możliwa praca z powietrzem olejonym (po rozpoczęciu olejenia trzeba je kontynuować)
Klasa odporności korozyjnej wg normy Festo	1 - niskie obciążenie korozyjne
Zgodność z LABS	VDMA24364-B2-L
Przydatność do pomieszczeń czystych, mierzona zgodnie z ISO 14644-14	Klasa 5 wg ISO 14644-1
Temperatura otoczenia	0 °C ... 80 °C
Energia uderzenia w pozycjach końcowych	0.7 J
Siła teoretyczna przy 0,6 MPa (6 bar, 87 psi), wycofanie	686 N
Siła teoretyczna przy 0,6 MPa (6 bar, 87 psi), przy wysuwie	754 N
Ruchoma masa przy skoku 0 mm	62 g
Dodatkowa poruszana masa na 10 mm skoku	9 g
Masa podstawowa przy 0 mm skoku	304 g
Dodatkowa masa na 10 mm skoku	45 g
Typ mocowania	Przy pomocy otworów przelotowych Przy pomocy gwintu wewnętrznego Przy pomocy osprzętu
Przyłącze pneumatyczne	M5
Informacja o materiałach	Zgodność z dyrektywą RoHS
Materiał pokrywy	Stop aluminium, anodowany
Materiał uszczelek dynamicznych	TPE-U(PU)
Materiał obudowy	Stop aluminium, anodowany
Materiał tłoczyska	Nierdzewna stal stopowa

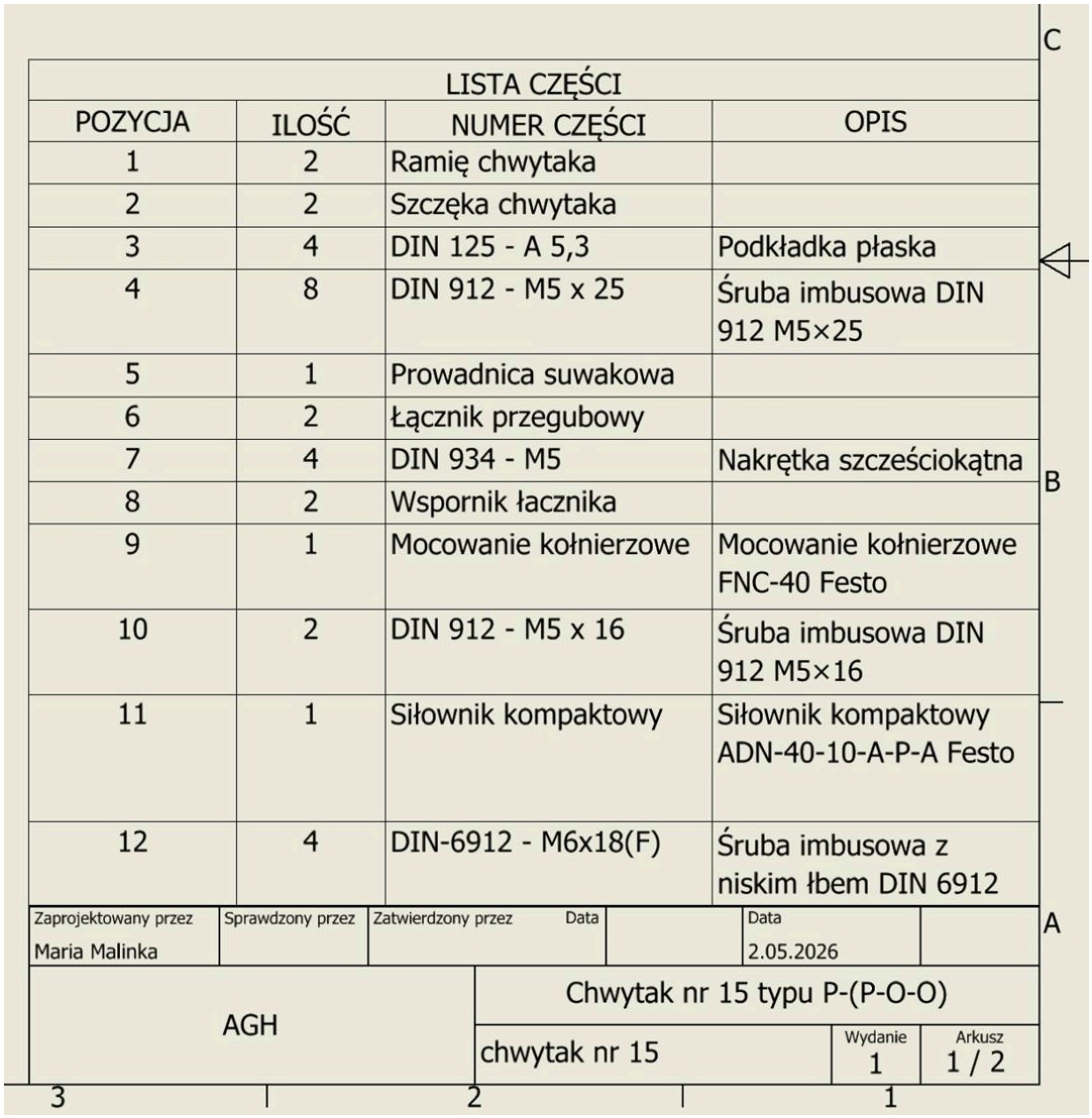
Key data from the manufacturer’s datasheet (translated):

Feature	**Value**
Stroke	10 mm
Piston diameter	40 mm
Cushioning	Elastic cushioning rings/pads at both ends
Mounting position	Any
Mode of operation	Double-acting

Feature	Value
Piston rod end	Male thread
Position sensing	For proximity switch
Variant	Single-ended piston rod
Operating pressure	0.06 MPa ... 1 MPa (0.6 bar ... 10 bar)
Operating medium	Compressed air to ISO 8573-1:2010 [7:4:4]
Ambient temperature	0 °C ... 80 °C
Impact energy at end positions	0.7 J
Theoretical force at 0.6 MPa (6 bar), retracting	686 N
Theoretical force at 0.6 MPa (6 bar), extending	754 N
Pneumatic connection	M5
Housing / cover material	Wrought aluminium alloy, anodised
Piston rod material	High-alloy stainless steel

12. Assembly drawing





LISTA CZĘŚCI

POZYCJA	ILOŚĆ	NUMER CZĘŚCI	OPIS
1	2	Ramię chwytaka	
2	2	Szczęka chwytaka	
3	4	DIN 125 - A 5,3	Podkładka płaska
4	8	DIN 912 - M5 x 25	Śruba imbusowa DIN 912 M5×25
5	1	Prowadnica suwakowa	
6	2	Łącznik przegubowy	
7	4	DIN 934 - M5	Nakrętka szcześciokątna
8	2	Wspornik łacznika	
9	1	Mocowanie kołnierzowe	Mocowanie kołnierzowe FNC-40 Festo
10	2	DIN 912 - M5 x 16	Śruba imbusowa DIN 912 M5×16
11	1	Siłownik kompaktowy	Siłownik kompaktowy ADN-40-10-A-P-A Festo
12	4	DIN-6912 - M6x18(F)	Śruba imbusowa z niskim łbem DIN 6912

Parts list (translated):

Item	Qty	Part number	Description
1	2	Gripper arm	
2	2	Gripper jaw	
3	4	DIN 125 – A 5.3	Flat washer
4	8	DIN 912 – M5 × 25	Hex socket head cap screw DIN 912 M5×25
5	1	Slide guide	
6	2	Pivot link	
7	4	DIN 934 – M5	Hex nut
8	2	Link bracket	
9	1	Flange mount	Flange mount FNC-40, Festo

Item	Qty	Part number	Description
10	2	DIN 912 – M5 × 16	Hex socket head cap screw DIN 912 M5×16
11	1	Compact cylinder	Compact cylinder ADN-40-10-A-P-A, Festo
12	4	DIN 6912 – M6 × 18 (F)	Low-head hex socket cap screw DIN 6912

Title block: designed by Maria Malinka, 2 May 2026, AGH – *Gripper No. 15, type P-(P-O-O)*, issue 1, sheet 1/2.

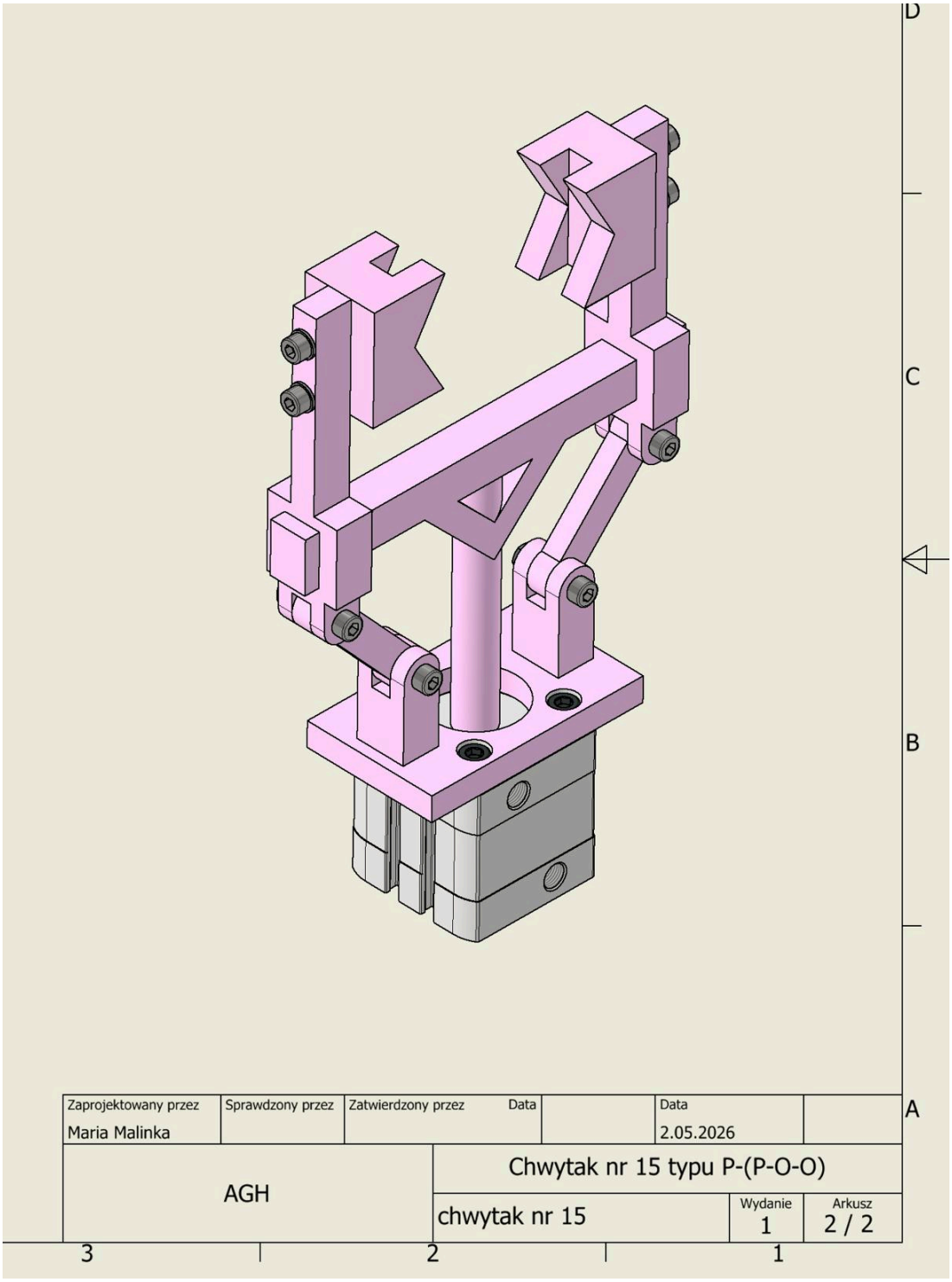


Fig. 13: Assembly drawing of the gripper created in Autodesk Inventor (isometric view, sheet 2/2)

13. 3D-printed prototype

To check the design in practice, the gripper was 3D printed in FDM technology on a Bambu Lab X1C printer from ASA material. The parts (arms, jaws, slide guide, links, brackets, and printed screws and nuts) were laid out on one build plate in Bambu Studio: long parts flat for better bed adhesion and strength, cylindrical parts and screws upright for more accurate diameters.

Parameter	**Value**
Printer / material	Bambu Lab X1C / ASA
Nozzle / layer height	0.4 mm / 0.2 mm
Nozzle / bed temperature	260 °C / 100 °C (Textured PEI plate, 5 mm brim)
Print time / material used	3 h 48 min / approx. 83.6 g

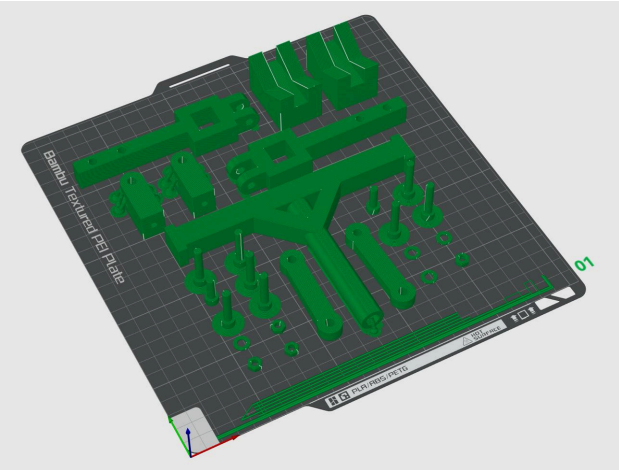


Fig. 14: Parts arranged on the build plate in Bambu Studio



Fig. 15: The finished 3D-printed gripper

The printed model reproduced the designed geometry with no significant warping, confirming that the gripper can be manufactured with FDM.